Emiliana Prates

Portugal | (+351) 911 049 764 | emilianaprates2005@gmail.com
LinkedIn: linkedin.com/in/emiliana-prates | Portfolio: vampxlia.github.io/portfolio/ | GitHub: github.com/vampxlia

PROFILE

Computer and Multimedia Engineering graduate and aspiring software engineer. I enjoy working across the stack and moving between different kinds of problems, from designing backend systems and web applications to building mobile experiences and experimenting with game development.

EDUCATION

BSc Computer and Multimedia Engineering

ISEL - Lisbon School of Engineering | Polytechnic University of Lisbon • Lisbon, Portugal • 08/2023 - 07/2026

Relevant Coursework: Algorithms & Programming, Object-Oriented Programming, Operating Systems, Computer Networks, Distributed Systems , Database Systems, Web Development, Mobile Applications, Artificial Intelligence, Machine Learning, Image Processing
Electives: Cybersecurity, Graphic Design, Interaction in Virtual Environments

Extracurriculars:
- ISEL Game Studios | Indie Game Developer | *Unity · Godot · C# · GDScript · Blender · Aseprite* • 05/2025 - 07/2026
- ISEL Space (AegiSat) | Embedded Systems Developer | *C++* • 08/2024 - 06/2025

PROJECTS

Course Certificate Generation System

02/2026 - 07/2026

TypeScript · Node.js · Express.js · MySQL · Docker · HTML · CSS

- Developed a full-stack application to automate certificate generation, digital signing and email delivery.
- Implemented role-based access control using Casbin and authentication with a MySQL database.
- Integrated academic databases to retrieve and validate student and course information.
- Implemented customizable certificate templates and cryptographic document signing.

View on Github - github.com/vampxlia/Course-Certificate-Generation-System

Ant Colony and Pheromone Simulator

11/2025 - 01/2026

Java · Processing · Agent-Based Simulation

- Developed an agent-based simulation modeling ant behavior.
- Implemented individual agent behavior and indirect communication through pheromone trails, producing emergent colony behavior.
- Built graphical visualization in Processing and integrated procedurally generated audio based on ant movement.

View on Github - github.com/vampxlia/Ant-Colony-and-Pheromone-Simulation

Arduino Gyroscope

10/2023 - 01/2025

C++ · Arduino · Embedded Systems

- Developed an embedded application to capture and display X/Y/Z accelerometer data in real time and convert it to tilt.
- Implemented I2C communication with an external LCD and serial data transmission over USB.
- Controlled the hardware directly without external sensor libraries.

View on Github - github.com/vampxlia/Arduino-Gyroscope

SKILLS

Tools & Frameworks: Node.js, Express.js, React.js, Spring, HTML, CSS, MySQL, SQLite, Git, Github, Docker
Programming Languages: Java, Typescript, Javascript, Python, Kotlin, GDScript, C#, C++, PHP
Additional Technologies: Arduino, Unity, Godot, Blender, Aseprite, Adobe Illustrator
Language Skills: Portuguese (Native), English (Fluent)